

Math 340 Practice Final

You are allowed a calculator and one 8.5" x 11" sheet of handwritten notes (both sides). The actual exam will have 14 questions. The Table of Laplace Transforms will be provided on the actual final. The practice exam is longer so you have a better sense of the types of problems that may be asked. Note that I often include a question on the actual exam that is not similar to questions on the practice exam.

Problem 1

Find all solutions to $\frac{dy}{dx} = e^{x-y}$

$$\frac{dy}{dx} = e^x e^{-y}$$

1st order
separable

$$\int e^y dy = \int e^x dx$$

$$e^y = e^x + C$$

or

$$y = \ln(e^x + C)$$

Problem 2

Solve the initial value problem $\frac{dy}{dx} = \frac{3x-2y}{2x+4y}$, $y(0) = 2$.

$$(2x+4y)dy = (3x-2y)dx$$

$$(2y-3x)dx + (2x+4y)dy$$

$$\frac{\partial M}{\partial y} = 2 = \frac{\partial N}{\partial x}$$

1st Order
Exact

$$\frac{\partial F}{\partial x} = 2y-3x \rightarrow F = \int 2y-3x \, dx = 2xy - \frac{3}{2}x^2 + C(y)$$

$$\frac{\partial F}{\partial y} = 2x+4y \rightarrow F = \int 2x+4y \, dy = 2xy + 2y^2 + C(x)$$

$$F = 2xy - \frac{3}{2}x^2 + 2y^2$$

Gen. Soln.

$$2xy - \frac{3}{2}x^2 + 2y^2 = K$$

Given $y(0)=2$ $2 \cdot 0 \cdot 2 - \frac{3}{2} \cdot 0^2 + 2 \cdot 2^2 = K$

$$8 = K$$

$$\boxed{2xy - \frac{3}{2}x^2 + 2y^2 = 8}$$

Problem 3

Find all solutions to $\frac{dy}{dx} + y = \frac{x}{y^2}$.

$$\frac{dy}{dx} + y = xy^{-2}$$

$$\frac{1}{3}v^{-2/3} \frac{dv}{dx} + v^{1/3} = xv^{-2/3}$$

$$\frac{dv}{dx} + 3v = 3x$$

1st order linear

$$\mu = e^{\int 3 dx} = e^{3x}$$

$$e^{3x} \frac{dv}{dx} + 3ve^{3x} = 3xe^{3x}$$

$$\frac{d}{dx} e^{3x} v = 3xe^{3x}$$

$$e^{3x} v = 3 \int x e^{3x} dx$$

$u = 3x \quad du = 3 dx$
 $dv = e^{3x} dx \quad v = \frac{1}{3} e^{3x}$

$$= 3 \left[\frac{1}{3} x e^{3x} - \int e^{3x} dx \right]$$

$$e^{3x} v = x e^{3x} - \frac{1}{3} e^{3x} + C$$

$$v = x - \frac{1}{3} + C e^{-3x}$$

1st order Bernoulli

$$\begin{cases} y = v^{\frac{1}{1-(-2)}} = v^{1/3} \\ \frac{dy}{dx} = \frac{1}{3} v^{-2/3} \frac{dv}{dx} \end{cases}$$

(Can solve either using 1st order linear OR linear constant coeff.)

linear constant coeff

Homog Soln
 $v_h = e^{-3x}$

Part. Soln.

Guess $v_p = Ax + B$

$$v_p' + 3v_p = A + 3Ax + 3B = 3x$$

$$A = 1$$

$$B = -\frac{1}{3}$$

$$v = x - \frac{1}{3} + C e^{-3x}$$

$$y = v^{1/3} = \left(x - \frac{1}{3} + C e^{-3x} \right)^{1/3}$$

Problem 4

Use the improved Euler's method with step size $h = 0.5$ to approximate $y(1.5)$ where

$$\frac{dy}{dx} = x^2 - y^2, y(0)=1.$$

x	y	$y'(x^2 - y^2)$	$\tilde{x}(=x+h)$	$\tilde{y}(=y+hy')$	$\tilde{y}'(=\tilde{x}^2 - \tilde{y}^2)$	$\frac{y' + \tilde{y}'}{2}$
0	1	-1	$\frac{1}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$
$\frac{1}{2}$	$\frac{3}{4}$	$-\frac{5}{16}$ (.3125)	1	$\frac{19}{32}$ (.59375)	$\frac{663}{1024}$ (.64746...)	$\frac{343}{2048}$ (.16748...)
1	.83371	.364277...	1.5	.9861788...	1.27745...	.79116...
1.5	1.229322					

$$y(1.5) \approx 1.229322$$

Problem 5

A tank has 10 kg of salt dissolved in 200 liters of water. Water with a salt concentration of 30g/liter flows into the tank at a rate of 2 liters/min, and the (perfectly mixed) water in the tank leaves at the same rate. How much salt is in the tank after an hour?

Let S = grams of salt in tank

$$\frac{dS}{dt} = \text{rate in} \times \text{concentration in} - \text{rate out} \times \text{concentration}$$

$$= 2 \text{ liters/min} \times 30 \frac{\text{g}}{\text{liter}} - 2 \text{ liter/min} \times \frac{S}{200} \frac{\text{g}}{\text{liter}}$$

$$= (60 - \frac{S}{100}) \text{ g/min}$$

$$\frac{dS}{dt} + \frac{S}{100} = 60 \quad S(0) = 10,000 \quad \leftarrow \text{Initially 10 kg}$$

$$\mu = e^{\int \frac{1}{100} dt} = e^{\frac{t}{100}}$$

$$e^{\frac{t}{100}} \frac{dS}{dt} + \frac{1}{100} e^{\frac{t}{100}} S = 60 e^{\frac{t}{100}}$$

NOTE: This equation is both separable and linear. If rate in $\neq$ rate out, it would be just linear, not separable

$$\frac{d}{dt} (e^{\frac{t}{100}} S) = 60 e^{\frac{t}{100}}$$

$$e^{\frac{t}{100}} S = \int 60 e^{\frac{t}{100}} dt = 6000 e^{\frac{t}{100}} + C$$

$$S = 6000 + C e^{-\frac{t}{100}}$$

$$S(0) = 6000 + C \stackrel{\text{Set}}{=} 10,000$$

$$C = 4000$$

$$S(t) = 6000 + 4000 e^{-\frac{t}{100}}$$

At 1 hour
 $t = 60 \text{ min}$

$$S(60) = 6000 + 4000 e^{-0.6} \approx 8195 \text{ g}$$

Problem 6

Solve the initial value problem $y'' + 6y' + 8y = 0$, $y(0) = 1, y'(0) = 1$.

Characteristic Eqn. $r^2 + 6r + 8 = 0$
 $(r+4)(r+2) = 0$
 $r = -2, -4$

$$y = C_1 e^{-2x} + C_2 e^{-4x}$$

$$y(0) = C_1 + C_2 \stackrel{\text{Set}}{=} 1 \quad \rightarrow \quad 2C_1 + 2C_2 = 2$$

$$y'(0) = -2C_1 - 4C_2 \stackrel{\text{Set}}{=} 1 \quad \rightarrow \quad -2C_1 - 4C_2 = 1$$

$$-2C_2 = 3$$

$$C_2 = -3/2$$

$$C_1 = 5/2$$

$$y = \frac{5}{2}e^{-2x} - \frac{3}{2}e^{-4x}$$

Problem 7

Find the general solution to $y'' + 9y = \cos(3x)$.

Characteristic Eqn. $r^2 + 9 = 0$
 $r = \pm 3i$

Complementary Soln.
(Soln. to Homo. Eqn.) $y = C_1 \cos(3x) + C_2 \sin(3x)$

Particular Soln. Since $\cos(3x)$ already in complementary soln., guess

$$y_p = Ax \cos(3x) + Bx \sin(3x)$$

$$y_p' = A \cos(3x) - 3Ax \sin(3x) + B \sin(3x) + 3Bx \cos(3x)$$

$$y_p'' = -3A \sin(3x) - 3A \sin(3x) - 9Ax \cos(3x) + 3B \cos(3x) + 3B \cos(3x) - 9Bx \sin(3x)$$

$$= -6A \sin(3x) - 9Ax \cos(3x) + 6B \cos(3x) - 9Bx \sin(3x)$$

$$y_p'' + 9y_p = -6A \sin(3x) - 9Ax \cos(3x) + 6B \cos(3x) - 9Bx \sin(3x) + 9Ax \cos(3x) + 9Bx \sin(3x)$$

$$= -6A \sin(3x) + 6B \cos(3x) \stackrel{\text{Set}}{=} \cos(3x)$$

$$A = 0, B = \frac{1}{6}$$

$$y_p = \frac{1}{6} x \sin(3x)$$

$$y = \frac{1}{6} x \sin(3x) + C_1 \cos(3x) + C_2 \sin(3x)$$

Problem 8

Write $4 \exp(i\pi/3)$ in rectangular form, $x + iy$.

$$\begin{aligned} 4e^{i\pi/3} &= 4(\cos(\pi/3) + i\sin(\pi/3)) \\ &= 4\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \\ &= \boxed{2 + i2\sqrt{3}} \end{aligned}$$

Problem 9

A mass of 2kg is attached to a spring, causing it to stretch 0.05m (5 centimeters). What amount of damping is critical for the spring-mass system? Use $g = 9.8\text{m/sec}^2$.

$$mg = k \Delta l$$

$$2 \cdot 9.8 = 0.05 \cdot k$$

$$k = 2 \cdot 20 \cdot 9.8 = 392 \text{ kg/sec}^2$$

Critical Damping when $C^2 = 4km$

$$C^2 = 4 \cdot 392 \cdot 2 \left(\frac{\text{kg}}{\text{sec}}\right)^2$$

$$C^2 = 3136 \text{ kg}^2/\text{sec}^2$$

$$C = 56 \text{ kg/sec}$$

Problem 10

Find the general solution to $x^2 y'' + 4xy' + 2y = 0$.

Euler Equation

Guess $y = x^r$

$$y' = r x^{r-1}$$

$$y'' = r(r-1)x^{r-2}$$

$$x^2 y'' + 4xy' + 2y = 0$$

$$r(r-1)x^r + 4rx^r + 2x^r = 0$$

$$(r^2 + 3r + 2)x^r = 0$$

$$r^2 + 3r + 2 = 0$$

$$(r+2)(r+1) = 0$$

$$r = -2 \text{ or } r = -1$$

$$y = C_1 x^{-2} + C_2 x^{-1}$$

Problem 11

Solve the initial value problem $x'' + 4x' + 5x = \sin(t^2)$, $x(0) = 0$, $x'(0) = 0$. Your answer will have an integral you can't simplify.

This can be done either by variation of parameters or Laplace transforms. Laplace transforms is probably easier for most people. Let $X = \mathcal{L}\{x\}$

$$s^2 X - s x(0) - x'(0) + 4(sX - x(0)) + 5X = \mathcal{L}\{\sin(t^2)\}$$
$$(s^2 + 4s + 5)X = \mathcal{L}\{\sin(t^2)\}$$

$$X = \frac{1}{(s+2)^2 + 1} \mathcal{L}\{\sin(t^2)\}$$
$$= \mathcal{L}\{e^{-2t} \sin(t)\} \mathcal{L}\{\sin(t^2)\}$$

$$X = \int_0^t e^{-2T} \sin(T) \sin((t-T)^2) dT$$

Problem 12

Solve the initial value problem $x'' + 2x' + 5x = \delta(t) + \delta(t-1)$, $x(0) = 0, x'(0) = 0$.

Laplace transform $\mathcal{L}\{x\} = X$

$$s^2 X - \overset{0}{s x(0)} - \overset{0}{x'(0)} + 2(sX - \overset{0}{x(0)}) + 5X = 1 + e^{-s}$$

$$(s^2 + 2s + 5)X = 1 + e^{-s}$$

$$X = \frac{1}{s^2 + 2s + 5} + \frac{e^{-s}}{s^2 + 2s + 5}$$

$$= \frac{1}{(s+1)^2 + 2^2} + \frac{e^{-s}}{(s+1)^2 + 2^2}$$

$$= \frac{1}{2} \frac{2}{(s+1)^2 + 2^2} + \frac{1}{2} \frac{2}{(s+1)^2 + 2^2} e^{-s}$$

So $\boxed{x = \frac{1}{2} e^{-t} \sin(2t) u(t) + \frac{1}{2} e^{-(t-1)} \sin(2(t-1)) u(t-1)}$

Problem 13

Find the a priori lower bound for the radius of convergence of the series solution to $(x^3 - 2x^2 - 3x - 20)y'' + xy' - 3y = 0$ about $x_0 = 0$. Hint: $x = 4$ is a singular point.

Singular points where $x^3 - 2x^2 - 3x - 20 = 0$
Since 4 is a singular point, we can factor out $x - 4$

$$\begin{array}{r} x^2 + 2x + 5 \\ x-4 \overline{) x^3 - 2x^2 - 3x - 20} \\ \underline{-(x^3 - 4x^2)} \\ 2x^2 - 3x \\ \underline{-(2x^2 - 8x)} \\ 5x - 20 \\ \underline{-(5x - 20)} \\ 0 \end{array}$$

$$(x^3 - 2x^2 - 3x - 20) = (x - 4)(x^2 + 2x + 5) = 0$$

$$x - 4 = 0 \Rightarrow x = 4$$

$$\begin{aligned} x^2 + 2x + 5 &= 0 \\ \Rightarrow x &= \frac{-2 \pm \sqrt{4 - 4 \cdot 5}}{2} \\ &= -1 \pm 2i \end{aligned}$$

Distance from 0 to 4 is 4

Distance from 0 to $-1 \pm 2i$ is $\sqrt{5}$ ✓ smallest

$$\boxed{\text{So R.O.C.} \geq \sqrt{5}}$$

Problem 14

Solve the initial value problem $y'' + (2x + 1)y' + y = 0$, $y(0) = 1$, $y'(0) = 1$. Write the series out to the x^4 term.

Since variable coefficient, look for a series solution

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad (2x+1)y' = \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Let $j = n-1$, $k = n-2$

$$y'' + (2x+1)y' + y = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k + \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j + \sum_{n=0}^{\infty} a_n x^n = 0$$

Collecting terms
(and changing all indices to n)

$$\cancel{n=0} \quad \cancel{n=1} \quad (2a_2 + a_1 + a_0) + \sum_{n=1}^{\infty} ((n+2)(n+1)a_{n+2} + 2na_n + (n+1)a_{n+1} + a_n) x^n = 0$$

$$a_2 = -\frac{a_1 + a_0}{2}$$

$$n \geq 1 \quad a_{n+2} = -\frac{(n+1)a_{n+1} + (n+1)a_n}{(n+2)(n+1)}$$

$$a_0 = y(0) = 1$$

$$a_1 = y'(0) = 1$$

$$a_2 = -\frac{2}{2} = -1$$

$$a_3 = -\frac{2a_2 + 3a_1}{6} = -\frac{1}{6}$$

$$a_4 = -\frac{3a_3 + 5a_2}{12} = \frac{11}{24}$$

$$y(x) = 1 + x - x^2 - \frac{1}{6}x^3 + \frac{11}{24}x^4 + \dots$$

Problem 15

Solve the initial value problem.

$$\frac{dx}{dt} = 4x + 4y, \quad x(0) = 4$$

$$\frac{dy}{dt} = x + 4y, \quad y(0) = 8$$

Take $\mathcal{L}$ of both sides of both equations $\frac{X - x(0)}{s} = \mathcal{L}\{y\}$

$$(s-4)\bar{X} = 4\bar{X} + 4\bar{Y}$$

$$(s-8)\bar{Y} = \bar{X} + 4\bar{Y}$$

$$(s-4)\bar{X} - 4\bar{Y} = 4 \rightarrow (s-4)\bar{X} - 4\bar{Y} = 4$$

$$-\bar{X} + (s-4)\bar{Y} = 8 \quad \begin{array}{l} -(s-4)\bar{X} + (s-4)^2\bar{Y} = 8(s-4) \\ \hline \end{array}$$

$$[(s-4)^2 - 4]\bar{Y} = 8s - 32 + 4$$

$$\bar{Y} = \frac{8s - 28}{s^2 - 8s + 12}$$

$$\frac{8s - 28}{s^2 - 8s + 12} = A \frac{1}{s-2} + B \frac{1}{s-6}$$

$$8s - 28 = A(s-6) + B(s-2)$$

$$s=2 \quad -12 = -4A \rightarrow A=3$$

$$s=6 \quad 20 = 4B \rightarrow B=5$$

$$\bar{Y} = 3 \frac{1}{s-2} + 5 \frac{1}{s-6}$$

$$y(t) = 3e^{+2t} + 5e^{+6t}$$

$$x(t) = -6e^{+2t} + 10e^{+6t}$$

$$(s-4)^2 \bar{X} - 4(s-4)\bar{Y} = 4(s-4)$$

$$-4\bar{X} + 4(s-4)\bar{Y} = 32$$

$$[(s-4)^2 - 4]\bar{X} = 4s + 16$$

$$\bar{X} = \frac{4s + 16}{s^2 - 8s + 12}$$

$$\frac{4s + 16}{s^2 - 8s + 12} = A \frac{1}{s-2} + B \frac{1}{s-6}$$

$$4s + 16 = A(s-6) + B(s-2)$$

$$s=2 \quad 24 = -4A \rightarrow A = -6$$

$$s=6 \quad 40 = 4B \rightarrow B = 10$$

$$\bar{X} = -6 \frac{1}{s-2} + 10 \frac{1}{s-6}$$

Problem 16

Solve the convolution equation for $x(t)$: $x(t) = \exp(-t) + \int_0^t \exp(t - \tau) x(\tau) d\tau$

$$x(t) = e^{-t} + e^t * x(t)$$

$$\underline{X} = \frac{1}{s+1} + \frac{1}{s-1} \underline{X}$$

$$\left(1 - \frac{1}{s-1}\right) \underline{X} = \frac{1}{s+1}$$

$$\frac{s-2}{s-1} \underline{X} = \frac{1}{s+1}$$

$$\underline{X} = \frac{s-1}{(s+1)(s-2)} = A \frac{1}{s+1} + B \frac{1}{s-2}$$

$$s-1 = A(s-2) + B(s+1)$$

$$s=2 \quad 1 = 3B \rightarrow B = \frac{1}{3}$$

$$s=-1 \quad -2 = -3A \rightarrow A = \frac{2}{3}$$

$$\boxed{x(t) = \frac{2}{3} e^{-t} + \frac{1}{3} e^{2t}}$$

Problem 17

Find and classify the critical points (equilibria) for the system

$$\frac{dx}{dt} = x^2 - y^2$$

$$\frac{dy}{dt} = x^2 + 2y^2 - 3$$

Find critical points

$$x^2 - y^2 = 0 \rightarrow x^2 = y^2 \text{ so } x = \pm y$$

$$x^2 + 2y^2 - 3 = 0 \rightarrow x^2 + 2x^2 = 3$$

$$3x^2 = 3$$

$$x^2 = 1$$

So critical points are $(\pm 1, \pm 1)$

Linearization

$$\begin{pmatrix} \frac{\partial}{\partial x}(x^2 - y^2) & \frac{\partial}{\partial y}(x^2 - y^2) \\ \frac{\partial}{\partial x}(x^2 + 2y^2 - 3) & \frac{\partial}{\partial y}(x^2 + 2y^2 - 3) \end{pmatrix} = \begin{pmatrix} 2x & -2y \\ 2x & 4y \end{pmatrix}$$

At $(-1, 1)$

$$\begin{pmatrix} -2 & -2 \\ -2 & 4 \end{pmatrix} \det = -12 < 0$$

Saddle

At $(1, 1)$

$$\begin{pmatrix} 2 & -2 \\ 2 & 4 \end{pmatrix} \det = 12 > 0$$

$$\text{trace} = 6 > 0$$

$$\text{tr}^2 - 4\det = -12 < 0$$

Spiral out

At $(-1, -1)$

$$\begin{pmatrix} -2 & 2 \\ -2 & -4 \end{pmatrix} \det = 12 > 0$$

$$\text{trace} = -6 < 0$$

$$\text{tr}^2 - 4\det = -12 < 0$$

Spiral in

At $(1, -1)$

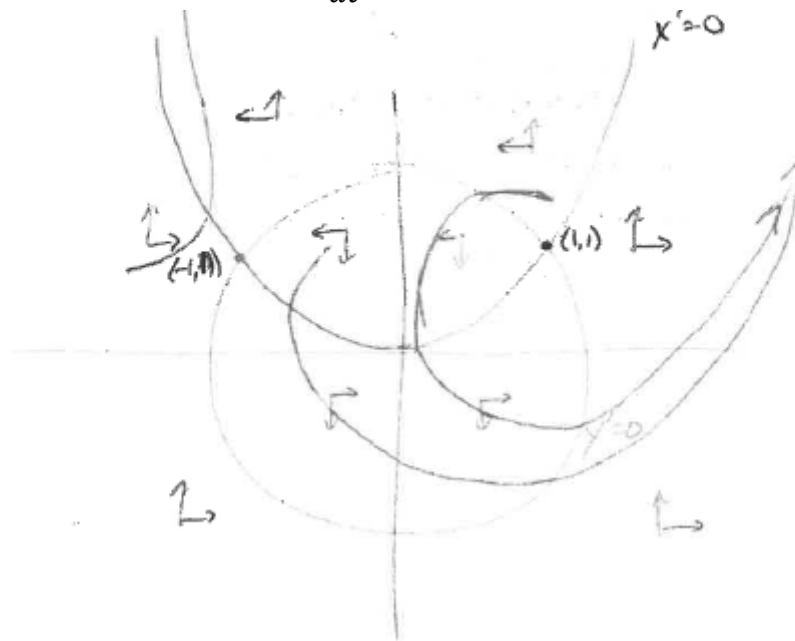
$$\begin{pmatrix} 2 & 2 \\ 2 & -4 \end{pmatrix} \det = -12 < 0$$

Saddle

Problem 18

Draw the curves $x' = 0$ and $y' = 0$ and sketch several (parametric) curves defined by the system

$$\begin{aligned}\frac{dx}{dt} &= x^2 - y \\ \frac{dy}{dt} &= x^2 + y^2 - 2\end{aligned}$$



Linearization

$$\begin{pmatrix} 2x & -1 \\ 2x & 2y \end{pmatrix}$$

At $(1, 1)$ $\begin{pmatrix} 2 & -1 \\ 2 & 2 \end{pmatrix}$ $\det = 6 > 0$ Spiral Out
 $\text{tr} = 4$
 $\text{tr}^2 - 4 \cdot \det = -8 < 0$

At $(-1, 1)$ $\begin{pmatrix} -2 & -1 \\ -2 & 2 \end{pmatrix}$ $\det = -6 < 0$ Saddle

Problem 19

Triton orbits Neptune with a semi-major axis of about $3.549 \times 10^8 m$ and orbital period about $5.078 \times 10^5 sec$. What is the approximate mass of Neptune?

Use $G \approx 6.67 \times 10^{-11} m^3 kg^{-1} sec^{-2}$ for the gravitational constant.

Since $T^2 = \frac{4\pi^2}{GM} a^3$ where $T = \text{orbital period}$
 $M = \text{central mass (mass of Neptune)}$

$a = \text{semi-major axis}$

$G = \text{gravitational constant}$

we have

$$M = \frac{4\pi^2 a^3}{GT^2} \approx 1.026 \times 10^{26} kg$$

Problem 20

A rocket is sent from the Earth (at 1 A.U.) to Mars (at 1.5 A.U.) along a Hohmann transfer orbit. How long will the trip take?

$$\text{Semi major axis} = \frac{1+1.5}{2} = 1.25 \text{ A.U.}$$

$$\text{Full } \text{period} \text{ orbit } T = (1.25)^{3/2} = 1.3975 \dots$$

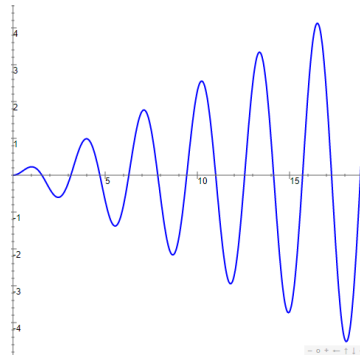
But to reach Mars we only need half the orbit

0.69877... years
or about 8.4 months

Problem 21

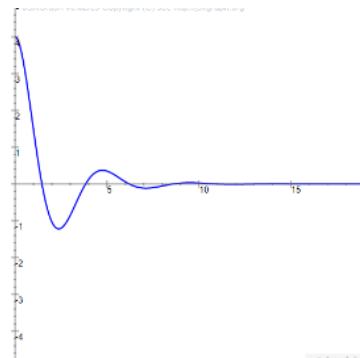
The following three equations each have a missing number. The three missing numbers are **1, 3, and 4**. Fill in the correct missing number for each equation in order to get the solution curve at the right of each equation. **You must fill in a different number for each equation.**

(a) $x'' + \underline{4} x = \cos 2t$



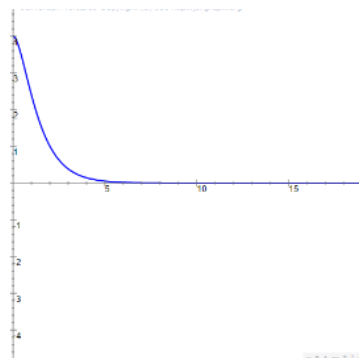
This is resonance.

(b) $x'' + \underline{1} x' + 2x = 0$



This is underdamped

(c) $x'' + \underline{3} x' + 2x = 0$



This is overdamped.